

POOJA KIRAN

SECURITY ENGINEER | AGENTIC AI SECURITY | CLOUD IAM

Tempe, AZ | +1 480-776-7745 | pkiran1@asu.edu | LinkedIn | GitHub | Portfolio

Technical Skills

Agentic & AI Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, Indirect Prompt Injection, RAG Security, LLM Red Teaming, Model Supply Chain Security, Adversarial ML

Application & Identity: Threat Modeling, Secure System Design, API Security, Capability-Based Authorization, Data Exfiltration Detection, AWS IAM, Workload Identity, Least Privilege, Trust Policies, Permission Boundaries

Engineering & DevSecOps: Python, Rust, C++, FastAPI, pytest, Hypothesis, GitHub Actions, CI/CD Security, SARIF 2.1.0, GitHub Code Scanning, CodeQL, Bandit, Trivy, pip-audit, Docker, Kubernetes

Detection & Observability: Elastic Security, ECS, SIEM, Prometheus, Security Telemetry, Tamper-Evident Audit Logging, Rate Limiting, MITRE ATLAS, OWASP LLM Top 10

Experience

Independent AI Security Researcher & Engineer

Aug 2024 - Present

Self-Directed Research | Self-employed

Tempe, AZ | Remote

- Built and evaluated security controls across **agent runtime security, AWS IAM, LLM red teaming, model supply-chain security, training-data integrity, and adversarial ML.**
- Engineered an **MCP/JSON-RPC security gateway with 55 prompt-injection patterns** and an **AWS IAM analyzer with 25 deterministic rules** covering agent tool calls, privilege escalation, trust-policy risk, and excessive permissions.
- Validated tooling with **659 passing MCP tests, 235 passing IAM tests, and 173 passing LLM security tests**, supported by CI, SARIF, GitHub Code Scanning, and security telemetry.

Business & Compliance Lead | AEROSSEC | Technology Innovation Lab Externship

Aug 2025 - Dec 2025

Honeywell Aerospace Technologies and Arizona State University

Tempe, AZ | On-site

- Led **business, compliance, and third-party security** work for AEROSSEC, a proposed protection layer for airline Passenger Service System integrations.
- Developed a **\$120K first-year commercialization scenario** and a **five-year financial model** covering revenue, operating costs, growth, and profitability; presented the case to ASU and Honeywell stakeholders.

Graduate Teaching Assistant | IT Grader

Jan 2025 - Oct 2025

Ira A. Fulton Schools of Engineering, Arizona State University | Part-time

Mesa, AZ | On-site

- Evaluated approximately **85 undergraduate web-development submissions per semester** for implementation quality, debugging, code correctness, and assignment requirements.
- Assessed **52 graduate cybersecurity assignments** covering security policies, compliance, risk analysis, and information-security controls; provided structured technical feedback with faculty.

Selected Projects

MCP Agent Security Gateway | Python, FastAPI, MCP/JSON-RPC, Elastic Security

Jul 2026 - Sep 2026

- Built an inline gateway that inspects routed agent tool calls using **55 prompt-injection patterns**, capability controls, PII/exfiltration signals, rate limiting, fail-closed behavior, and hash-chained audit logs.
- Validated **659 passing tests, 82% statement coverage, 9 Elastic Security rules, and 21 core SIEM tests.**

AWS Agent Identity Guard | Python, AWS IAM, SARIF 2.1.0

Aug 2026 - Sep 2026

- Built a static IAM analyzer with **25 deterministic rules** covering wildcard access, iam:PassRole, sts:AssumeRole, privilege escalation, trust-policy risks, audit tampering, and permission boundaries.
- Added **SARIF 2.1.0, GitHub Code Scanning compatibility, and CI exit-code enforcement**; validated **235 passing tests.**

HF Model Provenance Scanner | Python, SafeTensors, GGUF, ONNX, SARIF

Jul 2026 - Sep 2026

- Built a **non-executing model supply-chain scanner** using custom pickle-opcode analysis plus SafeTensors, GGUF, ONNX, Keras, AST/taint, provenance, and dependency checks.
- Detected **33/33 committed adversarial fixtures with 0 actionable findings across 4 benign samples**, explicitly scoped to the internal regression suite.

Education & Certifications

Arizona State University | M.S., Information Technology (Security), GPA: **3.87/4.00**

Aug 2024 - May 2026

M. S. Ramaiah University of Applied Sciences | B.Tech., CSE, CGPA: **8.44/10**

Aug 2019 - Aug 2023

AWS Academy: Cloud Architecting (Apr 2025) | Cloud Security Foundations (Nov 2025)